



Evaluating the economic and environmental benefits of deploying a national-scale, thermo-chemical plastic waste upcycling infrastructure in the United States

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ABSTRACT

Emerging chemical technologies can upcycle plastic waste by producing high-value polymers and other products. In this work, we study the economic and environmental benefits of deploying an upcycling infrastructure in the continental United States for producing low-density polyethylene (LDPE) and polypropylene (PP) from post-consumer plastic waste. Our analysis is based on a computational framework that integrates techno-economic analysis, life-cycle assessment, and value chain optimization. Our results demonstrate that the infrastructure could generate a market of nearly 20 billion USD per year and that this market is robust to various externalities. Our analysis also indicates that the infrastructure can achieve a plastic-to-plastic degree of circularity of 34% relative to residential plastic waste production, and leads to significant environmental benefits over alternative waste disposal methods, including 69%–75% lower greenhouse gas emissions than waste-to-energy systems and 38 million tonnes of avoided landfill waste per year.

1. Introduction

Global rates of plastic production, today reaching 400 million tonnes per year (Statista, 2022), require comprehensive and reliable plastic waste management systems (UNEP, 2021). While industry reports manufacturing process improvements (ACC, 2022), total plastic production and demand (particularly single-use plastics) continues to grow and overall plastic recycling rates remain well below 10% (Li et al., 2022). The bulk of plastic wastes are currently landfilled, incinerated, or allowed to accumulate in the environment (Hou et al., 2021). Historically, exportation has been a primary strategy among countries including the US to deal with waste plastic, but new limits imposed by waste-accepting countries (particularly China after 2018 (Katz, 2019)) have increased the urgency of domestic plastic waste management, and interest in alternative management strategies.

Traditional plastic recycling by mechanical methods is a mature pathway but faces significant inadequacies as a waste management strategy because it can only accept limited types of plastics (mainly bottles of polyethylene terephthalate-PET and high-density polyethylene-HDPE), which represent less than 12% of post-consumer plastic waste (Ma et al., 2023). Furthermore, mechanical recycling can yield lower quality products than its starting materials (and is therefore a form of “downcycling”) (Jehanno et al., 2022). In addition, mechanical

recycling methods cannot easily process flexible plastics, including multilayered films (e.g., used for food packaging), which represent roughly 35% of plastics currently produced (Xu et al., 2023). Ultimately, the lack of scalable recycling infrastructure promotes the need for waste disposal methods such as landfilling and waste-to-energy (WtE), which give rise to a host of environmental and social issues (e.g., land usage and greenhouse gas emissions).

An alternative strategy for plastic waste management, known as *upcycling*, uses chemical methods to decompose plastic polymers into constitutive monomers and precursor feedstocks that can be upgraded to obtain value-added products. Chemical upcycling pathways are more complex than mechanical recycling ones, as they involve complex sequences of processing steps; however, they can process a broader range of plastic waste streams. The chemical upcycling pathway that we analyze here is a thermo-chemical pathway that begins with the sorting and purification of plastic waste fractions of common recyclable waste streams at material recovery facilities (MRFs) and plastic reprocessing facilities (PRFs). Sorted plastic waste (in the form of mixed bales) undergo a series of chemical conversions including pyrolysis and steam cracking to arrive at pure plastic monomers (olefins), which can finally be repolymerized into products with the same original properties as virgin plastics (here, we target low-density polyethylene

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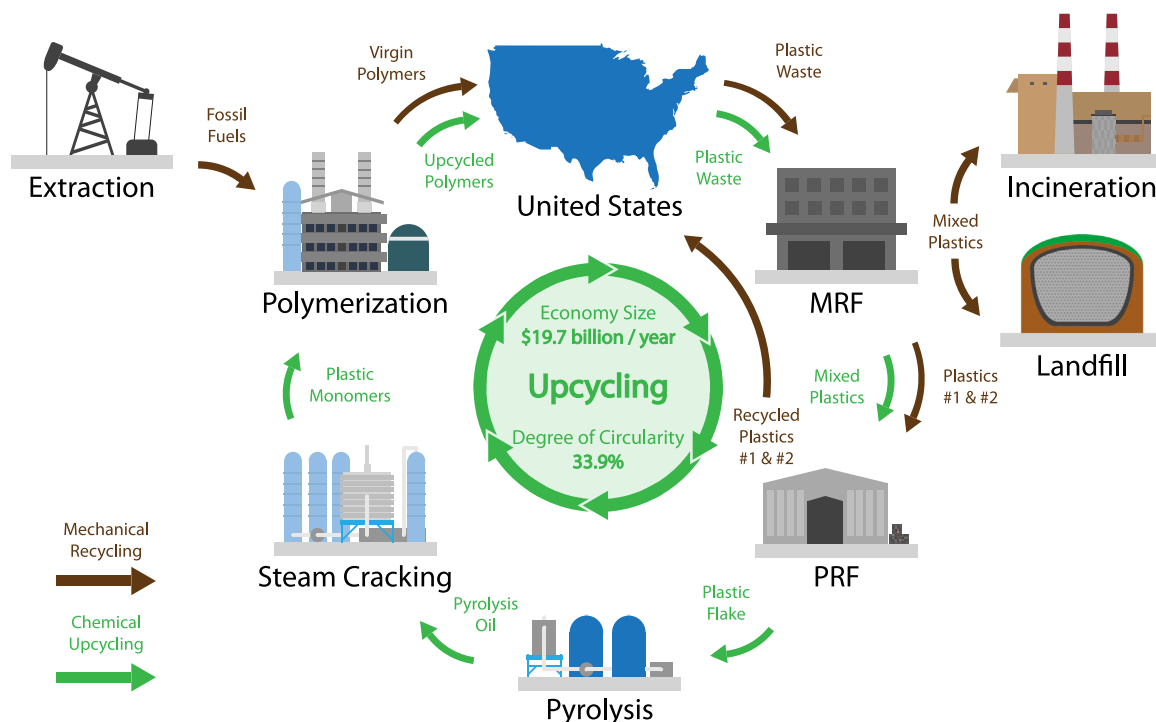


Fig. 1. Overview of the chemical upcycling value chain under study based on thermo-chemical processing. Chemical upcycling enables a circular economy where waste plastics are chemically reprocessed via a sequence of steps to obtain virgin-quality polymers. A combination of mechanical recycling for PET (plastic type #1) and HDPE (plastic type #2) and chemical upcycling of mixed plastics (plastic types, #3 - #7) can create an economy worth nearly 20 billion USD per year. When the network processes 100% of US residential plastic waste, it achieves a degree of plastic-to-plastics circularity reaching 33.9% (meaning that over one-third of input waste plastics return as mechanically or chemically recycled plastics) while a significant portion of the remaining inputs are converted into valuable byproduct chemicals.

and polypropylene, which are two of the most dominant plastics (Li et al., 2022)). The ability to recover original virgin plastics from mixed plastic waste enables the creation of a *circular plastics economy*.

Recent work has analyzed opportunities presented by upcycling and highlighted how diverse and multistep pathways can be integrated for potential contributions to plastic waste management (Li et al., 2022; Jehanno et al., 2022; Chen et al., 2021; Korley et al., 2021; Roy et al., 2021; Balu et al., 2022; Qian and Ren, 2024). Substantial work has also been devoted to analyzing the economic and environmental performance of various upcycling pathways (Hernández et al., 2023; Luo et al., 2023; Munguía-López et al., 2022), and additional work to the study of plastic waste distribution in geographical regions of interest (Milbrandt et al., 2022). However, limited work has explored logistical and interdependency issues that arise in the deployment of upcycling technologies in waste management infrastructure (e.g., how upcycling technologies connect with MRFs). Badejo et al. (2023) and Chen et al. (2023) proposed models for analyzing supply chains that manage plastic waste, but with a focus on polyethylene waste. Others have explored upcycling infrastructure optimization for smaller geographical scales; for instance, Cristiu et al. (2024) explored the optimization of pyrolysis infrastructure for Northern Italy in comparison to incineration alternatives, while Ma et al. (2023) proposed a supply chain optimization framework for studying the deployment of an upcycling infrastructure in the US Midwest region that processes post-consumer waste to obtain virgin LDPE and PP. This analysis revealed that economies-of-scale require infrastructures to be developed over sufficiently large geographical regions (to ensure sufficient supply of waste feedstock). Specifically, supply chain models are needed to capture complex trade-offs that arise from spatial distribution of waste, transportation costs and constraints, and technology sizes/locations.

In this work, we present a computational framework to examine the potential for deploying a national-scale upcycling infrastructure for post-consumer plastic waste in the continental US. The framework integrates techno-economic analysis, life cycle assessment (LCA), and

supply chain optimization in order to design an infrastructure that consists of a network of facilities (e.g., MRFs, PRFs, pyrolysis, and steam cracking) that are strategically sized and placed to capture post-consumer waste and obtain intermediate and final products to be delivered to consumers. The design aims to maximize total system profit (maximize value of derived products and minimize investment, processing, and transportation costs). We show that this infrastructure can be interpreted as a value chain (an economy) under which stakeholders exchange plastic waste and derived products to maximize profit.

2. Methodology

2.1. Computational framework

The proposed framework uses a mathematical optimization model (a large-scale, mixed-integer program-MIP) that aims to design a supply chain by strategically placing processing technologies that can capture post-consumer waste and obtain intermediate and final products that are delivered to consumers. This framework integrates data from techno-economic analysis and carbon footprint-focused LCA with supply chain optimization techniques. The optimization model can be interpreted as an infrastructure design model that aims to create an economy that maximizes total profit (which captures revenue obtained from all derived products and costs associated with investment, processing, and transportation costs) (Ma et al., 2023).

The model uses the US existing division into counties and county equivalents, where each county is localized to one node (its geographical center of population following recent US Census data (US Census Bureau, 2021a)) on a graph representation of the continental US. Nodes representing adjacent counties are connected by arcs which allow the transport of collected waste materials (recyclables) and processing intermediates (plastic bales, plastic flakes, and oil) around the

graph (US Census Bureau, 2021b). This graph representation of the US is presented in Fig. 2a.

Waste production is assumed to be directly related to population (it is estimated that each person in the US produces 130 kg of plastic waste each year) (Law et al., 2020). Municipal solid waste (MSW) composition statistics from the US Environmental Protection Agency (EPA) specify that approximately 23.7% of material in recyclable streams is plastic (US EPA, 2012), and over 80% of these plastics includes plastic types #3 - #7 and other lower-value types such as non-bottle PET and HDPE plastics (#1 - #2) (Ma et al., 2023). We apply these estimates to determine rates of recyclable material production across the continental US; for each county, total population is used to determine annual recyclable production, which is localized to its county node. The production of recyclable wastes is distributed across the US as in Fig. 2b, with higher production density linked to dense urban centers. See the supplementary information (SI) for further comments and data regarding US plastic waste availability.

2.2. Upcycling value chain structure

Our chemical upcycling pathway comprises six different processing technologies, including material recycling facilities (MRFs), plastic reprocessing facilities (PRFs), pyrolysis, steam cracking, LDPE polymerization, and PP polymerization. We focus on a pyrolysis-based form of chemical upcycling to build on previous work (Ma et al., 2023) and in recognition of the versatility of the process. The critical process output, pyrolysis oil, can be upgraded in a variety of ways. The subsequent processing steps we model (steam cracking and polymerization) prioritize the recovery of virgin-quality plastics from mixed plastics while also producing valuable byproduct fuels and chemicals.

The value chain involves 16 distinct products, including input products (post-consumer recyclables), intermediate products (e.g., plastic waste bales, plastic flake, pyrolysis oil, ethylene), and final products (e.g., LDPE, PP, fuels) (see S-Tables 4 & 6 for more details). Feedstock recyclables, plastic bales, plastic flake, and pyrolysis oil are allowed to be transported around the US by choosing realistic transportation costs, while the remaining (e.g., ethylene and propylene gas) involve high transportation costs (e.g., pipelines) that make transport infeasible (they need to be consumed on-site). These transportation restrictions are captured by the model. The techno-economic specifications of each component of the upcycling infrastructure are adapted from Ma et al. (2023) (which focused on the US Midwest region). See the SI for details on technologies and for further details on process yield factors, capital costs, and operating costs.

Residential waste producers/sources across the US provide mixed recyclable streams, where plastics are mixed with other recyclable materials (e.g., paper, metal, and glass). Our model assumes that residents are capable of sorting recyclables from other MSW without significant contamination in the disposal process, and that all MRFs are newly deployed and capable of handling the sorting of mixed plastics. MRFs separate plastics from other recyclables and the remaining recyclables are no longer considered in the scope of our infrastructure (because there are well-developed systems in place to deal with these materials). Excluding the recycling of paper, metal, and glass from our analysis does not detract from our economic results, but could in fact underestimate the total value of the infrastructure. In addition, our modeling framework can be easily extended to account for the recycling of these materials. Similarly, emission estimates of upcycling account for the sorting and processing functions of the MRF and PRF stages, but not the additional processing of excluded paper, metal, and glass streams. The model assumes that HDPE and PET bottles are separated at MRFs and are mechanically recycled; this pathway is considered within the scope of the infrastructure. The remaining plastic waste (#3 - #7) is recovered in MRFs in the form of mixed plastic bales that are reprocessed (e.g., clean up) and fed to pyrolysis processes. As analyzed by Ma et al. (2023), it is possible to consider the scenario of feeding HDPE and PET to pyrolysis systems; in this work, however, we aim to understand how chemical upcycling can complement existing mechanical recycling infrastructure.

2.3. Life cycle assessment (LCA)

Basic LCA methodologies were used to estimate the environmental impacts of a given process or product across several diverse categories. In this analysis, our goal is to estimate and compare the carbon footprints of upcycling, landfilling, and waste-to-energy (WtE) within an LCA framework. We acknowledge that carbon footprint carries limitations as a single impact category, and to achieve a more comprehensive environmental assessment, an analysis of multiple environmental LCA categories would be required (Arfelli et al., 2024; Laurent et al., 2012; Vlasopoulos et al., 2023). However, we focus on carbon footprint for this analysis as it is a primary environmental indicator in the chemical upcycling industry and literature, and because process uncertainties in these emerging technologies make other impact categories more challenging to estimate.

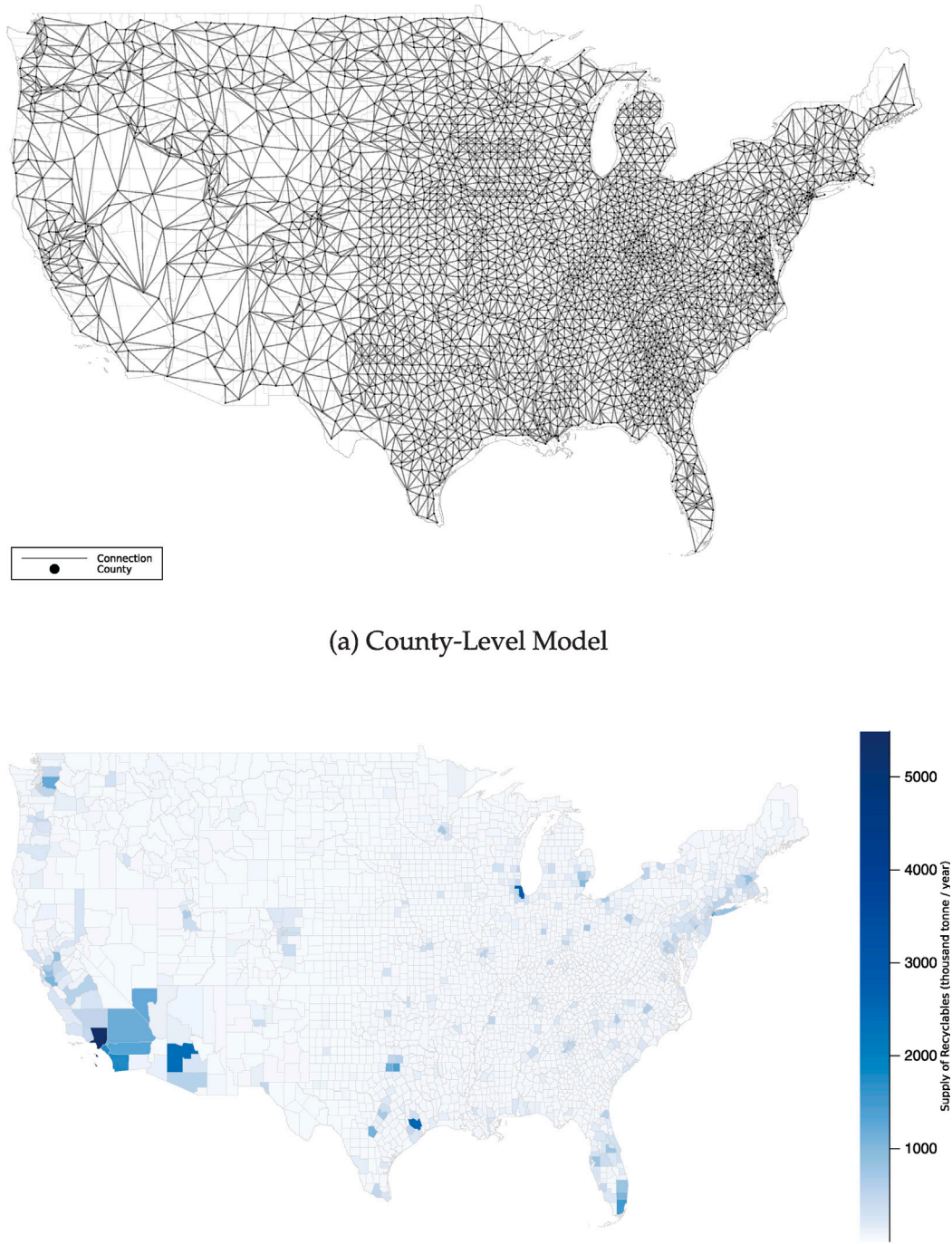
We define the scope of our study to include all process steps in the upcycling network, from the collection of mixed waste plastics from households to the production of LDPE and PP virgin-grade polymers (summarized in Fig. 1). We also account for the transportation of recyclables, plastic bales, plastic flakes, and pyrolysis oil. See S-Figure 11 for a system boundary diagram.

Critical greenhouse gas (GHG) emission factors for each technology in the upcycling sequence are presented in Fig. 3 (also detailed in S-Table 7). These estimates are adapted from Ma et al. (2023), which quantified global warming potential of process GHG emissions on a 100-year time horizon (GWP100) using TRACI impact categories. All associated product and technology specifications were adapted from literature reports, industry sources, and engineering insight. For the emissions analysis, the global warming impacts of each technology comprise natural gas for heating applications and electricity from the US average national grid (US EIA, 2023), while fuel use in product transportation is accounted for separately. Emission factors for each technology are computed with a basis of 1 kg of the technology reference input product, described in Fig. 3. Comparative GHG emissions factors for landfilling and WtE are obtained from the Waste Reduction Model (WARM), a US EPA tool that estimates high-level emissions of various waste management practices (US EPA, 2023).

We present direct, unadjusted emissions comparisons, as well as comparisons with emissions credits applied to waste management strategies that create additional valuable products. All methods fulfill the purpose of accepting post-consumer plastic waste, but upcycling produces valuable polymers, and WtE produces electricity which would otherwise be sourced from the grid. Emissions factors for replaced grid electricity are estimated from a national grid mix of fuels (US EIA, 2023), and GHG emissions for each fuel according to the EcoInvent database (ecoinvent, 2024) used with SimaPro LCA software (PRé Sustainability, 2024). Emission factors for replaced virgin products are likewise estimated with LCA SimaPro software and EcoInvent database, where each product is approximated as a pre-defined database product equivalent reported in Table 1). Each database equivalent possesses a region and other specifications that introduce some uncertainty into our emission crediting calculations, which can be refined in future work. We present estimates (with and without emission credits for avoided products) on a functional unit basis of 1 kg of mixed plastic waste processed. Further information on upcycling system boundaries and emissions estimates can be found in the SI.

2.4. Circularity

A central objective of upcycling is to enable a circular economy for plastics, in which currently unrecyclable plastics avoid disposal by being converted to valuable fuel or chemical byproducts, or by being recovered as virgin-quality plastics. While conversion extends the lifecycle of virgin plastics, recovery as new plastics is more truly circular, as the constituent chemicals continue to be used as plastics.



(a) County-Level Model

(b) National Recyclables Production

Fig. 2. (a) County-level graph/network representation of the US, which is comprised of 3108 nodes and 9164 distinct connection (transportation) arcs. (b) Based on population counts from the 2020 US Census, our model estimates a total supply of 181 million tonnes of recyclable materials (42.8 million tonnes of which are plastics that are processed through the upcycling infrastructure), which is distributed into county-level totals.

Thus, we discuss the circularity of the upcycling infrastructure using two metrics.

We define overall carbon degree of circularity as a ratio of valuable product mass to input feedstock mass, as in Eq. (1). This metric represents how well upcycling minimizes material process losses. Valuable

products include all plastics as well as fuel and chemical byproducts which are sold and contribute value to the total process revenue.

$$\text{overall carbon circularity} = \frac{\text{mass of valuable products}}{\text{mass of input plastics}} \quad (1)$$

In contrast, our defined plastic-to-plastic degree of circularity metric is more restrictive, accounting only for the portion of plastic process

Table 1

Product identities from SimaPro 9.1.0.7 with EcoInvent database used to determine appropriate emission credits toward upcycling operations. Emission credits are applied towards a couple of main polymer products (LDPE and PP) and five byproducts. See S-Tables 2 and 6 for full product definitions.

Product	Database equivalent	Lifecycle emissions (kg CO ₂ /kg)	Emissions credit (kg CO ₂ /kg plastic)
LDPE	Polyethylene low density granulate (PE-LD), production mix, at plant RER	2.095	0.291
PP	Polypropylene, PP, granulate, at plant/RER	1.619	0.136
Pyrolysis Gasoline	Pyrolysis gasoline, at plant/kg/RNA	1.546	0.183
Wax	Slack wax, at plant, US SE/US	0.537	0.073
Fuel Gas	Refinery gas {GLO} market for APOS, U	0.573	0.051
Fuel Oil	Light fuel oil {RoW} light fuel oil production, petroleum refinery operation APOS, U	0.441	0.017
C ₄	N-olefins {GLO} market for APOS, U	1.963	0.128

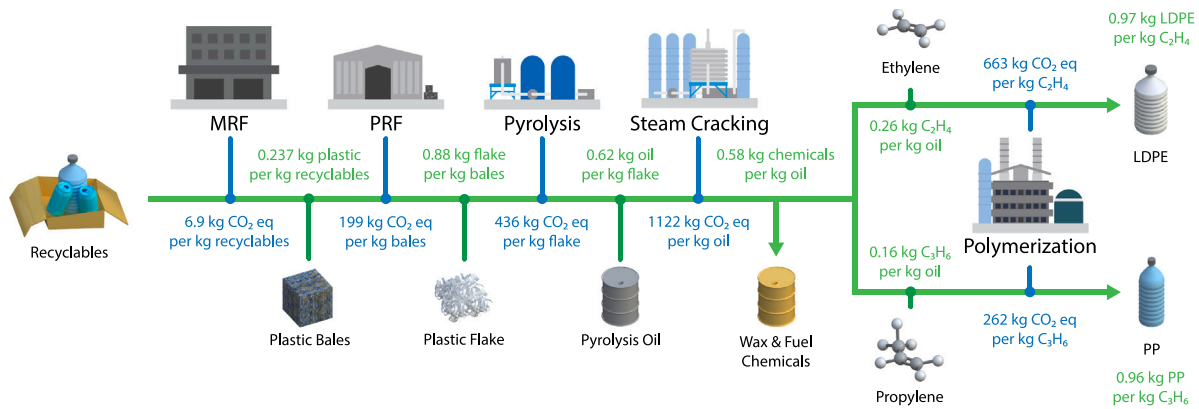


Fig. 3. Product conversion yield factors and GHG emissions coefficients for upcycling technologies. MRF and PRF conversion factors are based on a national approximate MSW plastic fraction and distribution of plastic types. In this analysis, PET and HDPE bottles are sorted out of mixed plastics streams through a combination of MRF and PRF operation for separate mechanical recycling and recovery of PET and HDPE, so the conversion from plastic bale to plastic flake reflects the diversion of mechanically recyclable plastics. See S-Tables 2 & 7 for further details on technology yield factors and emission factors.

outputs relative to input feedstock mass, as in Eq. (2). This metric represents how well the process prioritizes plastic products and avoids byproducts that do not remain in the plastics economy.

$$\text{plastic-to-plastic circularity} = \frac{\text{mass of plastic products}}{\text{mass of input plastics}} \quad (2)$$

2.5. Optimization model

The value chain optimization model uses a mixed-integer programming (MIP) formulation, originally proposed by Ma et al. (2023) to study the potential for a plastic upcycling network in the US Midwest region. An MIP formulation allows us to identify the optimal placement, number, and specifications for all processing facilities in our defined upcycling network, while also tracking the movement of supplied products, intermediates, and final products in space and their contributions to system value.

Given the extremely large size of the MIP for the national-scale system that we study here, we conducted some simplifications to enable computational tractability. Specifically, transportation routes are limited to adjacent counties and we limit the size of technologies to a couple of options. These simplifications significantly reduce model complexity and solution time, but the scale of the problem still represents a challenge to state-of-the-art solvers. As such, here we compute close-to-optimal solutions and report the optimality gap of such solutions (which measures how far the solution is from the calculated upper bound on total profit) to justify that our designs are of high enough quality. Future work will aim to implement advanced

algorithms that enable the solution of problems with a higher level of resolution and complexity.

The model captures product supply, demand, transportation and transformation conducted by a collection of stakeholders in the value chain, including waste providers, waste transporters, waste processors, and end-product consumers. We assign notation to the properties of the above four stakeholder classes of the value chain as follows. County waste providers (residents) are collectively indexed in the set $i \in S$ each with a supplying cost $\alpha_i^s \in \mathbb{R}_+$, maximum supplying capacity $\bar{s}_i \in \mathbb{R}_+$, and the amount of plastic supplied, also referred to plastic supply flow $s_i \in \mathbb{R}_+$. The upcycling facility network is represented as a set of technology providers $m \in \mathcal{M}$ with processing costs $\alpha_m^e \in \mathbb{R}_+$ (which include contributions from annualized capital costs normalized by plant capacity), processing capacity $\bar{\xi}_m \in \mathbb{R}_+$, and the amount of product processed $\xi_m \in \mathbb{R}_+$. We model the value chain for upcycling end products as consumers present every county node (indexed as $j \in D$) with arbitrarily high capacities ($\bar{d}_j \in \mathbb{R}_+$), uniform product demanding price ($\alpha_j^d \in \mathbb{R}_+$), and the amount of product demanded, referred to as demand flow ($d_j \in \mathbb{R}_+$). Finally, the transporters of recyclables, plastic wastes, and other upcycling intermediates are indexed by the set $l \in \mathcal{L}$, with transportation cost $\alpha_l^t \in \mathbb{R}_+$, capacities $\bar{f}_l \in \mathbb{R}_+$, and the amount of product moved $f_l \in \mathbb{R}_+$. We use subsets to simplify the notation for transport: $\mathcal{L}_{n,p}^{in}$ is the subset of transport providers $l \in \mathcal{L}$ who deliver product p to node n , and similarly $\mathcal{L}_{n,p}^{out}$ are those who transport p away from n . We use a similar notation for product transformation with the set $\mathcal{M}_{n,p}^{gen}$ used to represent those technologies producing a product p at node n and $\mathcal{M}_{n,p}^{con}$ those technologies consuming it. The parameters $\gamma_{m,p}$ are the input and output yield coefficients for each technology (e.g., the

mass of LDPE that can be produced from a given basis of polyethylene monomers).

Using these parameters and variables, the optimization problem is formulated as in (3).

$$\max_{s,d,f,\xi} \sum_{j \in D} \alpha_j^d d_j - \sum_{i \in S} \alpha_i^s s_i - \sum_{l \in L} \alpha_l^f f_l - \sum_{m \in M} \alpha_m^{\xi} \xi_m \quad (3a)$$

$$\text{s.t.} \quad \sum_{i \in S_{n,p}} s_i + \sum_{l \in L_{n,p}^{\text{in}}} f_l + \sum_{m \in M_{n,p}^{\text{gen}}} \gamma_{m,p} \xi_m = \sum_{j \in D_{n,p}} d_j + \sum_{l \in L_{n,p}^{\text{out}}} f_l + \sum_{m \in M_{n,p}^{\text{con}}} \gamma_{m,p} \xi_m, \quad (n, p) \in \mathcal{N} \times \mathcal{P}, (\pi_{n,p}) \quad (3b)$$

$$0 \leq s_i \leq \bar{s}_i, \quad i \in S \quad (3c)$$

$$0 \leq d_j \leq \bar{d}_j, \quad j \in D \quad (3d)$$

$$0 \leq f_l \leq \bar{f}_l, \quad l \in L \quad (3e)$$

$$0 \leq \xi_m \leq y_m \cdot \bar{\xi}_m, \quad m \in M \quad (3f)$$

The objective (3a) is to maximize the total system profit (also known as economic surplus), which is given by the total demand revenue and the total supply costs, transformation costs (including operating and annualized capital), and transport costs. The constraint (3b) involves balances at all geographical locations $n \in \mathcal{N}$ and for each product $p \in \mathcal{P}$. Constraints (3c) through (3f) are capacity limits for demand, supply, transport, and transformation. Constraint (3f) incorporates binary variables y_m ; these variables indicate whether the facility is deployed at the given location. If $y_m = 0$, the facility is not placed, and processing capacity is zero; otherwise, $y_m = 1$ and the facility is installed and its capacity is available.

In our case study, demand capacity limits are set arbitrarily high so that all products of the upcycling value chain can be consumed. These demand flows d_j are used to determine the total amounts of upcycling polymer products (chemically upcycled virgin LDPE and PP) and traditionally recycled products (mechanically recycled HDPE and PET) formed in each case study. The demand flows of all valuable products are compared to the sum of supply capacities $\bar{s}_i$ to calculate the overall carbon degree of circularity for the upcycling network (Eq. (1)), while the demand flows for LDPE, PP, rPET, and rHDPE only are used to calculate the network's plastic-to-plastic circularity (Eq. (2)). Finally, upcycling revenue is determined as a sum of products between the demand flows for each product d_j and the demanding price for each product α_j^d .

The solution of the MIP presented above identifies the optimal binary variables y_m (the optimal design of the system that determines technology types, sizes, and locations). Following this optimization, we then fix these optimal binary variables in a distinct second stage of modeling to obtain a linear programming (LP) problem; the solution of the LP is equivalent to assume that the technologies are already deployed and thus we seek to find the best management of a value chain (best allocation of products between stakeholders) to maximize total profit. From the solution of the LP we obtain the dual variable of the product balance constraints $\pi_{n,p}$; these values are key and are interpreted as the clearing price of product p at the location node n . The clearing price $\pi_{n,p}$ is the inherent value of the product p at geographical location n . As such, the model helps discover the inherent value for all products in the value chain (e.g., recyclables, pyrolysis oil, plastic bales); these prices vary in space, because they implicitly capture transportation costs and regional availability of waste. These prices also provide the mechanism to compensate stakeholders involved in the value chain and help us understand how revenue is distributed among stakeholders. For instance, even if counties offer recyclables for

free, plastic upcycling can generate significant value from waste, which could be used to compensate county residents for providing their waste (as well as technologies and transportation for providing their services). The prices obtained by the model are such that they cover the operating costs of all stakeholders; for more details on the economic properties of the value chain model, please refer to the work of Tominac and Zavala (2021).

3. Results

3.1. Computational analysis

The optimization model of the national upcycling problem comprises 4010 geographical locations that are available as candidates for technology placement, each with a defined technology type and a couple of sizes (capacities) which can be selected (with capital costs that scale nonlinearly with capacity to capture economies of scale). In total, the MIP problem contains 796,756 continuous variables, 8020 binary variables, and 587,405 constraints; the large number of variables (which capture potential technology selection and placement decisions) make the problem challenging to solve.

The total number of sites was determined based on the minimum number of smallest-scale technologies of each type required to allow full waste capture (totaling 2005), multiplied by a buffer factor to not overly restrict the optimized solution. These sites were assigned to county nodes in order of descending population (meaning that the largest counties received technology sites first), in order to facilitate the placement of facilities near population hubs. This procedure limits the number of binary variables needed to formulate the problem.

To further reduce model size, we connected only adjacent counties with arcs to allow product flow, visualized in Figure 2a. This arrangement uses the minimum number of arcs to fully connect all 3108 county nodes (9164 adjacent connections compared to 3108^2 possible connections), while eliminating trivial arcs that would add significantly to model size but minimally to model solution accuracy (such as connections between two small and distant counties). These sites were assigned to county nodes in order of descending population (meaning that the largest counties received technology sites first), in order to facilitate the placement of facilities near population hubs. This procedure limits the number of binary variables needed to formulate the problem. To further reduce model size, we connected only adjacent counties with arcs to allow product flow. This arrangement uses the minimum number of arcs to fully connect all county nodes, while eliminating trivial arcs that would add significantly to model size but minimally to model solution accuracy (such as connections between two small and distant counties).

We implemented the MIPs in the algebraic modeling package JuMP (Bezanson et al., 2017) and solved them using Gurobi 9.1.0. (Gurobi Optimization, LLC, 2022). The MIPs were used to design the layout of the upcycling value chain, while its economic and environmental properties are analyzed using the **CoordinatedSupplyChains.jl** package (CSC.jl) developed by Tominac and Zavala (2021) and modified for this work with visualization capabilities. The CSC.jl framework relies on the same conception of a value chain with four distinct categories of stakeholders, and the same mathematical optimization problem as in (3), with the only difference that the binary technology placement variables y_m are fixed to values determined in solving the MIP. The CSC.jl analysis package is publicly available at <https://github.com/ui/Packages/CoordinatedSupplyChains>. All scripts and data needed to reproduce the results are available at https://github.com/zavalab/JuliaBox/tree/master/National_Plastic.

The optimal (baseline) infrastructure design for a continental US upcycling network identified with our framework is presented in Fig. 4. After 180 h of computational time, the MIP was solved to an optimality gap of 4.66%, meaning that the identified system design provides an economic result within 5% of the optimal solution. The optimal design

reaches all 3108 counties and captures 100% of residential plastic waste; in other words, the model indicates that there are sufficient economic incentives to harness all the post-consumer waste to obtain virgin plastics (no policy intervention is needed). A couple of technology placement patterns are evident in the optimal design: facilities involved in the low-level (inexpensive) sorting and processing steps (MRFs and PRFs) are distributed, while facilities performing higher-level (expensive) chemical processing (pyrolysis, steam cracking, and polymerization) are more centralized (as these strongly benefit from economies of scale). We also observe the formation of seven distinct “hubs” in the US that are strategically placed to harness all waste (while minimizing transportation costs). A summary of the technologies selected are summarized in Table 3. Here, we can see that 336 MRFs are deployed, while only a handful of steam cracking units are deployed. The model can also be easily modified to restrict the number and location of steam cracking facilities (which would lead to a suboptimal solution), but these additional scenarios are not explored (to keep the discussion simple).

3.2. Circularity & economic evaluation

By capturing 100% of residential waste, the network produces a combination of mechanically-recycled and chemically-upcycled plastic products, whose volumes are summarized in Table 2. In total, 11.6% of residential waste plastics are recovered as mechanically-recycled HDPE and PET through MRFs and PRFs (from a limited feedstock of PET and HDPE bottles), and 22.3% are recovered chemically as LDPE and PP through pyrolysis, steam cracking, and repolymerization (from the remaining mixed plastic portion of the feedstock). This leads to a plastic-to-plastic degree of circularity of 33.9%, relative to the total rates of residential plastic waste production.

As discussed in the Methodology section, it is important to note that this upcycling plastic-to-plastic degree of circularity is limited due to inherent byproducts obtained in pyrolysis and steam cracking processing (e.g., carbon embedded in waste plastics ends in byproducts and not in plastics). This is an important limitation of the upcycling pathway based on thermo-chemical technologies. However, including these valuable byproducts, the upcycling infrastructure has an overall carbon circularity of 94.6%, demonstrating low material losses and efficient use of input feedstock. Critically, this assumes no losses of mixed plastic waste feedstock in the collection or MRF sorting stages before pyrolysis and subsequent technologies.

It is also important to note that though a lower mass of plastic products are recovered through upcycling than mechanical recycling, known process degradation prevents mechanical recycling from being truly circular. Its products have lower quality and fewer applications than upcycling products. However, our model combines these two strategies to show that upcycling can recover additional plastics from mixed plastic wastes that would normally be ineligible for mechanical recycling, and thus, the processing strategies are complementary.

We also determined that the recovered plastic products of the infrastructure lead to a volume reduction of 64.3% of residential mixed plastic waste (this estimate uses average densities of individual resins and a US distribution of resins in mixed plastic waste (Vitz et al., 2016; Statista, 2022)). This large volume reduction is an important benefit over landfilling. The results in Table 2 also highlight how LDPE and PP are significantly more valuable polymers than HDPE and PET; as such, there is an economic incentive to process waste HDPE/PET to obtain LDPE/PP via thermo-chemical upcycling (as opposed to using mechanical recycling).

Analysis of prices obtained by the model indicates that thermo-chemical upcycling can make plastic waste and other intermediate products highly valuable and indicates that such values vary substantially by geographical location, as illustrated in Fig. 5 and Table 4. Importantly, the plastic waste value is positive everywhere in the continental US, revealing a key economic result for the infrastructure:

the upcycling infrastructure is economically viable without added incentives (policy). Its viability is derived from value created in part by the reclamation of mechanically-recycled HDPE/PET plastics (Table 2), but the majority of the system value arises from the production of high-value LDPE/PP through thermo-economical upcycling. The model also reveals that upcycling creates an economy worth 19.7 billion USD annually (a measure of total economic surplus, excluding capital costs). This total profit can be used to remunerate all stakeholders, including producers of recyclable streams (residents). For instance, if residential profit is 10 billion USD per year, this would represent an average per-county profit of 3.3 million USD/year and an average per-capita profit of 31 USD/year. Simply put, upcycling provides an opportunity for US residents to be compensated for recycling plastic (or to not be charged tipping fees), which could provide incentives for correct residential-level sorting of waste plastic. We also observe that residential profits are strongly affected by geographical location; as Fig. 5 reveals, counties closest to MRFs and processing hubs benefit the most, as they avoid transportation costs. This economic behavior captures the fact that waste in remote areas and with low population densities is more difficult and expensive to capture.

3.3. Uncertainty/sensitivity analysis

The economic performance of the upcycling infrastructure is sensitive to various technical specifications and value chain externalities such as waste offering cost for recyclables, transport costs, polymer market values, and investment budgets available to cover capital costs. In the baseline solution represented by Fig. 4, US residential waste producers donate recyclables (offered for free) and capital investment budgets are unconstrained, while transport costs, polymer market values, and technology efficiencies remain unchanged from the analysis of Ma et al. (2023). Here, we compare performance of the upcycling infrastructure by running additional model case studies with adjustments to these parameters; we measure impact of these parameters in terms of the percent of total waste captured by the upcycling infrastructure and the total profit of the system (economic surplus).

Fig. 6 demonstrates that the upcycling infrastructure is robust to diverse economic scenarios. For instance, we can see that the system remains viable even if residents offer their waste at high values (residents would like to get paid for providing the waste feedstock). Specifically, we find that total waste captured remains high (>99%) for offering costs of up to 40 USD/tonne (but economic surplus decreases). Waste capture drops off sharply for higher offering costs.

Fig. 6 also shows the performance of the upcycling infrastructure with adjustments to the efficiencies of critical technologies (including conversion of plastic flake to pyrolysis oil at pyrolysis facilities, and the conversion of pyrolysis oil to ethylene and propylene monomers at steam cracking facilities). When technology efficiency is increased, the economic surplus significantly increases (as these conversions are key precursor steps to the formation of valuable end products). However, we also note that, as efficiencies increase, the total waste captured decreases slightly in some cases; this is because, as efficiency increases, more product is formed and the constrained upcycling infrastructure reaches capacity faster (upcycling is able to involve fewer residential providers while producing better overall economics).

Fig. 6 demonstrates the performance of upcycling when transportation costs are all raised by a uniform percentage increase. Our analysis indicate that transportation costs can be doubled and the system will still capture 99.47% of all residential plastic waste and the economic surplus only decreases slightly; this indicates that transportation costs are not major drivers in the economics of the system (a counterintuitive result).

Our analysis shows that the market values for the end-product polymers (LDPE and PP) are the main economic drivers for upcycling infrastructure viability. S-Figure 9 in the SI reports the levels of waste captured for different combinations of market values for the polymers,

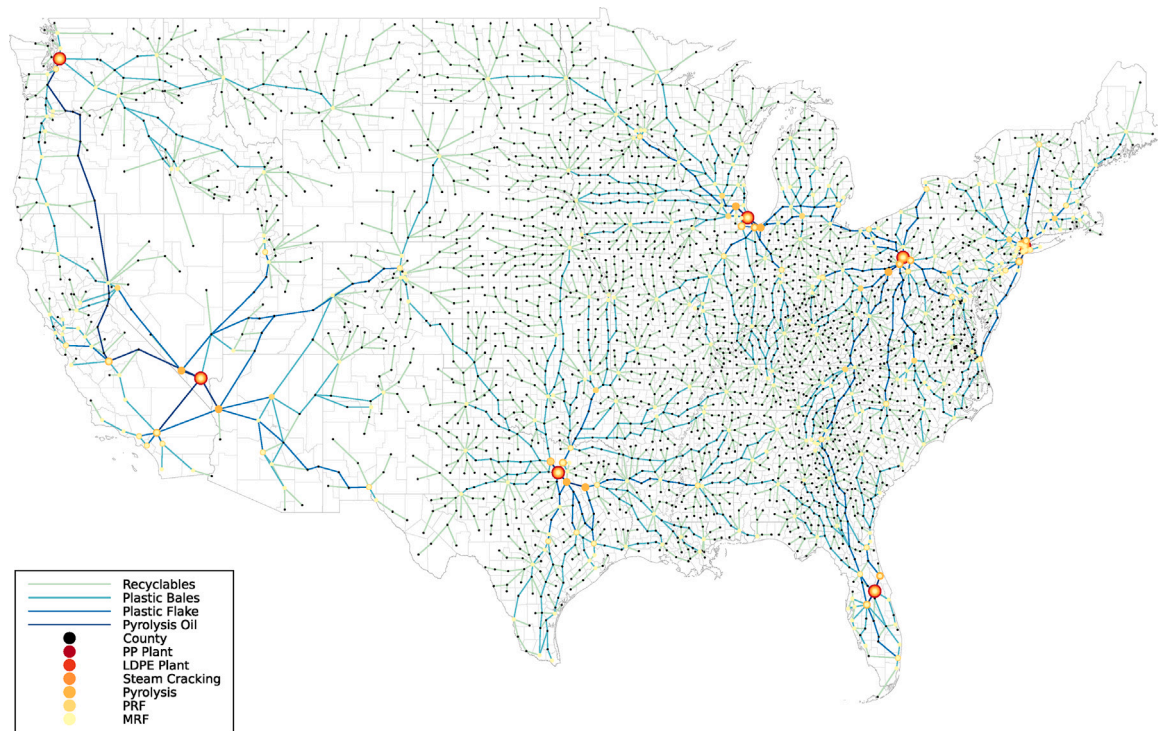


Fig. 4. Optimal upcycling infrastructure, showing the placement of all the involved technologies. This map also shows the transport of county recyclables to nearby MRFs, and the transport/exchange of key intermediate products between technologies.

Table 2

Summary of key model solution statistics for recyclable and plastic waste production, mechanical-recycled products (HDPE/PET), and upcycling products (LDPE/PP).

Product	Recyclables supplied	Plastic supplied	HDPE produced	PET produced	LDPE produced	PP produced
Quantity (tonne/year)	180,683,328	42,803,880	2,054,590	2,910,660	5,953,880	3,595,230
Volume (m ³ /year)	–	41,218,170	2,162,726	2,132,351	6,471,609	3,950,802
Avg Value (USD/tonne)	50.34	335.48	991.00	418.00	2204.00	2424.00

Table 3

Summary of key technology facility statistics. Total capacity used is a weighted average of used capacity measured for each type of technology, and total capacity in units of plastic bales is back-calculated using this weighted average and the total amount of plastic waste in the model. This demonstrates that the value chain is not at capacity and could accept further plastic waste, suggesting opportunities for incorporating other sources of waste (e.g. commercial & industrial) or accommodating population growth.

Technology	# Deployed	Reference input product	Total capacity (tonnes reference)	% Capacity used
MRF	336	Recyclables	222,640,000	81.15%
PRF	64	Plastic Bales	45,200,000	94.70%
Pyrolysis Plant	34	Plastic Flake	38,690,000	97.80%
Steam Cracking	7	Pyrolysis Oil	23,880,000	97.96%
LDPE Plant	12	Ethylene	7,200,000	85.25%
PP Plant	7	Propylene	4,800,000	78.02%
Total	460	Plastic Bales	50,600,000	84.58%

Table 4

Spread of prices for recyclables and three early intermediates in the upcycling value chain.

Product	Recyclables	Plastic bales	Plastic flake	Pyrolysis oil
Minimum Value (USD/tonne)	18.73	284.28	435.95	696.60
Average Value (USD/tonne)	50.34	335.48	470.85	740.35
Maximum Value (USD/tonne)	72.18	395.24	548.41	831.24

measured in decrements of 10% from the baseline market values. Fig. 6 reports market viability at four sample combinations where the value of both polymers are reduced by the same percentage: total waste captured decreases minimally even with a 20% decrease in polymer values, but this decrease does reduce upcycling profitability. Furthermore, we find that the market value of PP is more critical to upcycling performance, as it is the dominant (valuable) polymer product over LDPE.

The total capital cost (upfront investment) of the base upcycling infrastructure design shown in Fig. 4 was 51.4 billion USD. We investigated the effects of having limited investment budgets by resolving the MIP with total capital costs constrained by percent decrements from the base solution value. This involves adding constraints on the annualized capital cost component of the sum of processing costs α_m^ε in (3). The results are shown in Fig. 6; despite the relative linearity of the relationship between percent of total waste captured in upcycling and capital cost, we highlight that limiting capital investment widens

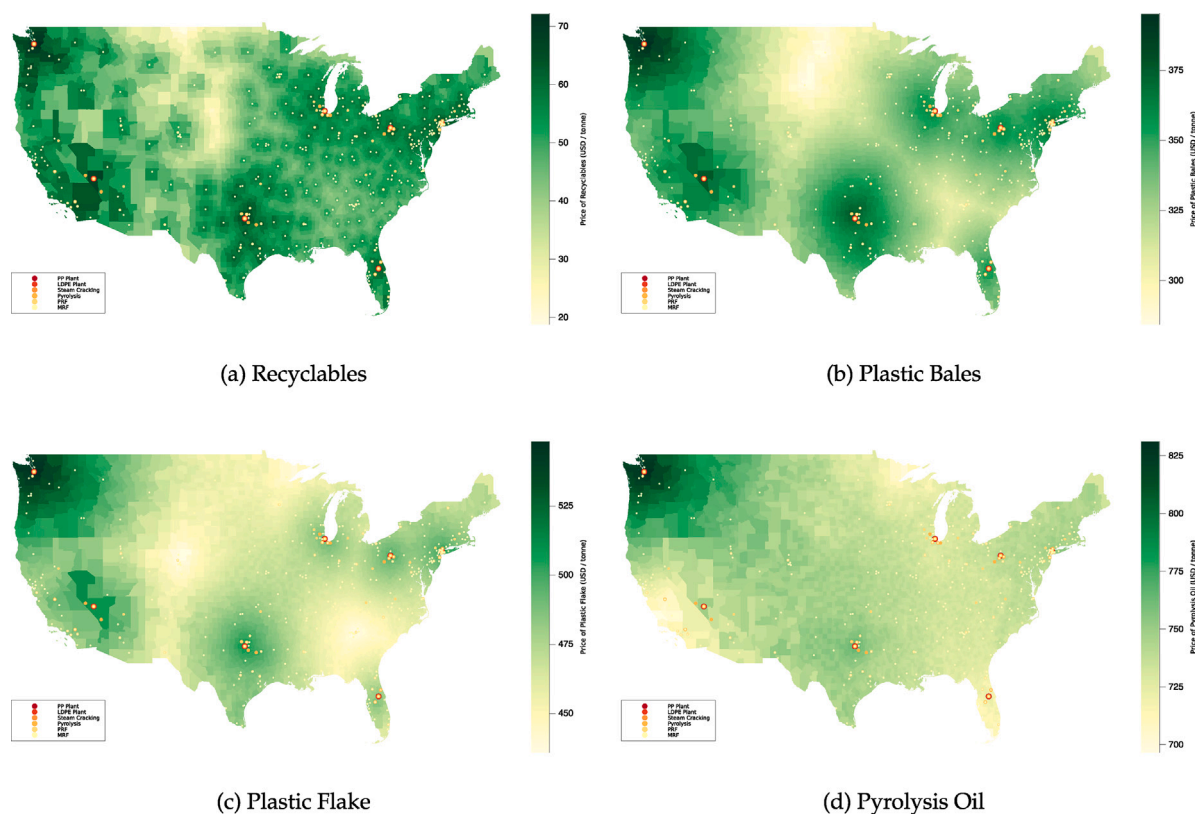


Fig. 5. Inherent values (prices) of recyclables and intermediates in the upcycling value chain. The value of starting recyclables is highest in close proximity to distributed MRFs and centralized processing hubs, but greater transportation distances and costs detract value created by upcycling.

processing disparities at the regional level. As shown in S-Figure 10 in the SI, the optimal upcycling network when investment is limited prioritizes areas of higher population density and higher waste production (such as the urban centers of the East and West coasts), but largely excludes regions such as the Great Plains and Northwest. This analysis thus reveals how the system naturally prioritizes regions with the highest economic potential.

Previous work examining upcycling on the regional scale for the US Midwest (Ma et al., 2023) provides an interesting point of comparison for this national-scale analysis. For the analogous case which allows PET and HDPE to be processed by traditional methods, Ma et al. determined that the inherent value of mixed plastic waste is in the range of 336–493 USD/tonne, whereas in our study we find plastic waste having a lower value range of 284–395 USD/tonne. With all processing costs equivalent, this comparison suggests that transportation costs detract more significantly from plastic waste value in the national value chain solution. This is in part due to the deployment of fewer and more centralized processing technologies, which may be desirable from a capital cost minimization perspective. Such considerations invoke an important assumption of our modeling approach, which designs and infrastructure “from scratch” (without accounting for existing infrastructure). In a real implementation of upcycling, existing US MRFs would likely be incorporated, contributing to a deployment pattern based on regional needs rather than a nationally optimized framework. However, obtaining MRF data is difficult and complicates analysis.

3.4. Comparison with existing processing methods

3.4.1. Direct impacts

A potential upcycling infrastructure for the continental US diverts plastic which would normally become waste to the chemical production of virgin-quality polymers. As such, upcycling reduces the volume of

waste input to existing landfill or WtE systems, with economic and environmental consequences.

Key economic costs of landfills and WtE systems avoided by upcycling are their tipping fees, assessed on a per-mass basis for waste disposed at their facilities. For landfills, a national average tipping fee reaches 61.01 USD/tonne of material (Katner and Staley, 2020), and tipping fees for WtE run higher (Castaldi et al., 2022). In comparison, the cost of processing plastic through upcycling far exceeds these tipping fees at 232.63 USD/tonne (excluding the operating costs of MRFs, which are also required for waste sorting in landfill and WtE scenarios). However, even if chemical upcycling is more expensive (as it involves more complicated processing), the value of LDPE/PP polymer products formed through this chain more than compensates such costs. This means that none of the processing costs have to be passed on to residential waste producers for the system to be economically viable, unlike landfill and WtE disposal systems which rely on tipping fees (residents lose money in such systems). Assuming that PET/HDPE in an upcycling implementation continue to be mechanically recycled due to existing market support, there is a significant economic savings in not disposing of the mixed plastic fraction of recyclables (totaling nearly 38 million tonnes annually) at landfills or WtE plants, which exceeds 2.3 billion USD/year.

Diverting plastic from landfills also has environmental impacts with regard to land consumption. We analyze the extent of US landfill infrastructure using data from the EPA’s Landfill Methane Outreach Program (LMOP) (US EPA, OAR, 2022). From the subset of US landfills which are currently open and operational, we estimate the average waste capacity and land area of a US landfill at 22 million tonnes and 180 acres, respectively. Extending this estimate to the quantity of mixed plastic waste generated annually in the US, the implementation of upcycling could save over 300 acres of land per year from landfill consumption. Over a ten year period, the diversion of plastic waste into upcycling could avoid the construction of 18 new landfills—over

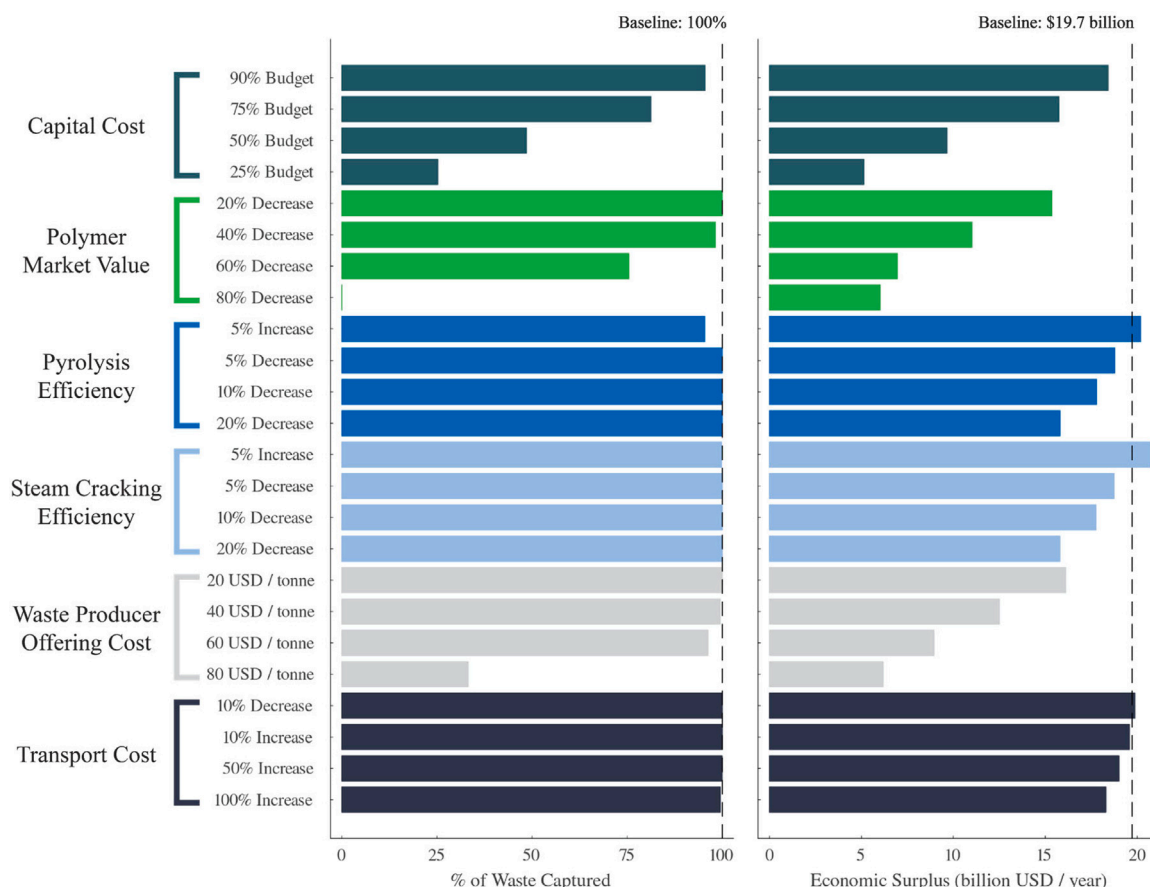


Fig. 6. The effectiveness and economic performance of the upcycling infrastructure are functions of model setup parameters including capital cost budgets, polymer market values, technological efficiencies, waste producer offering prices, and transport costs. Variation in fraction of waste captured by upcycling (left) and economic surplus (right) is here reported with adjustments to these four parameters individually. All waste producer offering prices are measured in USD/tonne of recyclables. Constraining capital costs requires resolving the MIP and obtaining new value chain designs (samples of which are presented in S-Figure 10), but all other parameters are adjusted using the same network solution in Fig. 4.

20 years, this becomes 35 landfills, and over 50 years, 86 landfills. Using this estimate of land consumption, we can also apply a range of capital costs for new landfill construction, taken to be between 300,000–800,000 USD/acre (Duffy, 2016). Thus, upcycling also avoids between 90–240 million USD in landfill capital expenses annually. Depending on the assumed lifespan of the upcycling network, avoided landfill construction costs could reach from 1.8–4.8 billion USD (for a 20 year lifespan) to 4.5–12 billion USD (for a 50 year lifespan). This does not include separate costs of landfill closure and capping, estimated at 227,000–326,000 USD/acre (Duffy, 2005). Therefore, upcycling avoids an additional 68.1–97.8 million USD/year in landfill closing costs, or 3.4–4.9 billion USD over 50 years.

A more limited analysis can be performed assuming that upcycling instead diverts plastic from WtE systems, because land consumption is not a primary concern for WtE, and the US WtE network is not nearly as expansive as its network of landfills. Applying data from the Energy Recovery Council's 2018 inventory of WtE in the US, the average annual capacity for accepting waste of a US WtE plant is 410,000 tonnes/year (Michaels and Krishnan, 2018). This means that over 100 WtE plants of this size would be required to process all residential plastic waste from the continental US. This is unrealistic, as only 74 WtE commercial plants existed in the continental US in 2018 (and several have closed since then (Redling, 2020; Rosengren, 2018)), and the existing facilities are concentrated in certain regions (e.g. the Northeast).

The total GHG emissions of the upcycling value chain (including MRF activity) are 1.345 tonne CO₂ eq/tonne of plastic waste processed. Transportation of recyclables and relevant upcycling intermediates contributes an additional 0.042 tonne CO₂ eq equivalent per

tonne of plastic processed, resulting in overall emissions of 1.387 tonne CO₂ eq/tonne of plastic. In contrast, landfills produce minimal direct GHG emissions from plastic degradation, but may produce emissions through landfill operation. Using the EPA WARM model (US EPA, 2023), we estimated the impact of landfilling mixed plastic wastes (including landfill operation and an assumed transportation distance of 20 miles from households to landfills) in terms of diesel fuel consumption, and then determined a GHG emission factor from this consumption rate using the EcoInvent database. The resulting landfill emission factor is estimated at 0.0236 tonne CO₂ eq/tonne of plastic, which is significantly lower than that of upcycling. However, it must be acknowledged that plastic disposal in landfills has significant negative environmental impacts not considered when focusing only on GHG emissions, including land use and water and land pollution via microplastics. Other environmental benefits of upcycling over landfill systems fall under the category of land consumption. Finally, direct WtE emissions range from 2.3–3.1 tonnes CO₂ eq per tonne of plastic combusted, depending on resin type. Contributions to emissions estimates include the transportation of plastic waste to the facility, ash byproduct disposal, emission of non-biotic CO₂, and release of N₂O.

Direct emissions comparisons between all three waste management summarized are presented in Fig. 7, with additional economic and capacity comparisons. Choosing to incinerate mixed plastics in WtE rather than using a landfill would result in an increase in direct GHG emissions between 2.2–3.0 tonnes CO₂ eq per tonne of plastic disposed, while upcycling presents a direct GHG emissions savings relative to WtE of 0.9–1.7 tonnes CO₂ eq per tonne of plastic. In the US, mixed plastic wastes are typically processed by a combination of landfilling

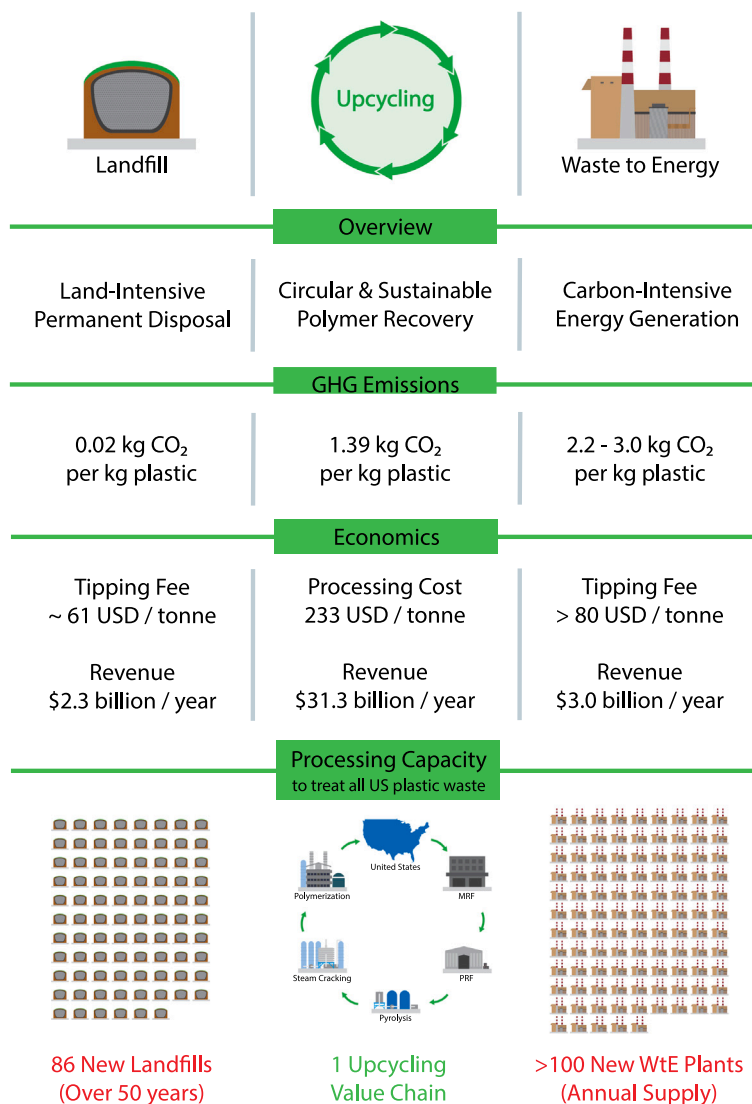


Fig. 7. Comparison between a national upcycling infrastructure and existing processing methods of landfilling and WtE. Revenue of upcycling is derived from the model solution, whereas revenue from landfilling and WtE pathways are based on tipping fees. Revenue from mechanical recycling of PET and HDPE is excluded, as are MRF operation costs, as both contribute to all three scenarios.

and WtE that varies by region. Thus, the true environmental benefits of implementing upcycling would lie at some intermediate point between the two extremes of landfill-only and WtE-only disposal; that is, upcycling would realize partial benefits in reducing WtE emissions and other benefits in avoiding land consumption.

3.4.2. Credited impacts

Upcycling and WtE fulfill additional functions relative to other waste management systems: upcycling also produces valuable plastic polymer products, and WtE produces electricity from waste plastic. In contrast, landfill systems offer a disposition route without creating any additional valuable products.

Any efforts to accurately compare emissions between these waste management avenues must take into account the impacts of virgin polymer production from fossil fuels which upcycling avoids by chemically recovering these polymers from waste, as well as the avoided fossil fuel consumption from WtE-generated electricity. For this purpose, we employ LCA methods and expand the system boundaries to account for benefits of replacing currently produced products (i.e., plastics) in the US economy in the case of the upcycling framework and produced electricity in the case of the WtE framework.

We credit upcycling for the production of valuable polymers LDPE and PP that would otherwise be manufactured from virgin fossil fuels, and for the production of other valuable byproducts (including pyrolysis gasoline, wax, fuel gas, fuel oil, and C₄ streams, see Ma et al. (2023) for further product inventory details). Removing these carbon-intensive processes reduces the upcycling emissions factor to 0.508 tonne CO₂ eq/tonne plastic (including transport).

WtE facilities generate energy in addition to managing plastic waste; as such, by similar reasoning, its corrected rate of emissions must take into account credits for avoided energy generation from fossil fuels. Net WtE emissions, accounting for energy generation credits, range from 1.6–2.0 tonnes CO₂ eq per tonne of plastic combusted depending on plastic type, which is significantly greater than the emissions of plastic input to upcycling. However, WtE would only receive GHG emissions credits as long as the electricity mix is sufficiently fossil-fuel based. Current credits were calculated using a national average grid electricity mix with an emission factor of 0.51 tonne CO₂ per MWh. As the grid becomes decarbonized, these GHG emission credits will vanish as grid electricity would already be produced from renewable sources.

Our emissions comparison with incineration technologies is presented in Fig. 8. Given that the impacts of incinerating plastics vary slightly based on resin type, we present incineration impacts, credits,

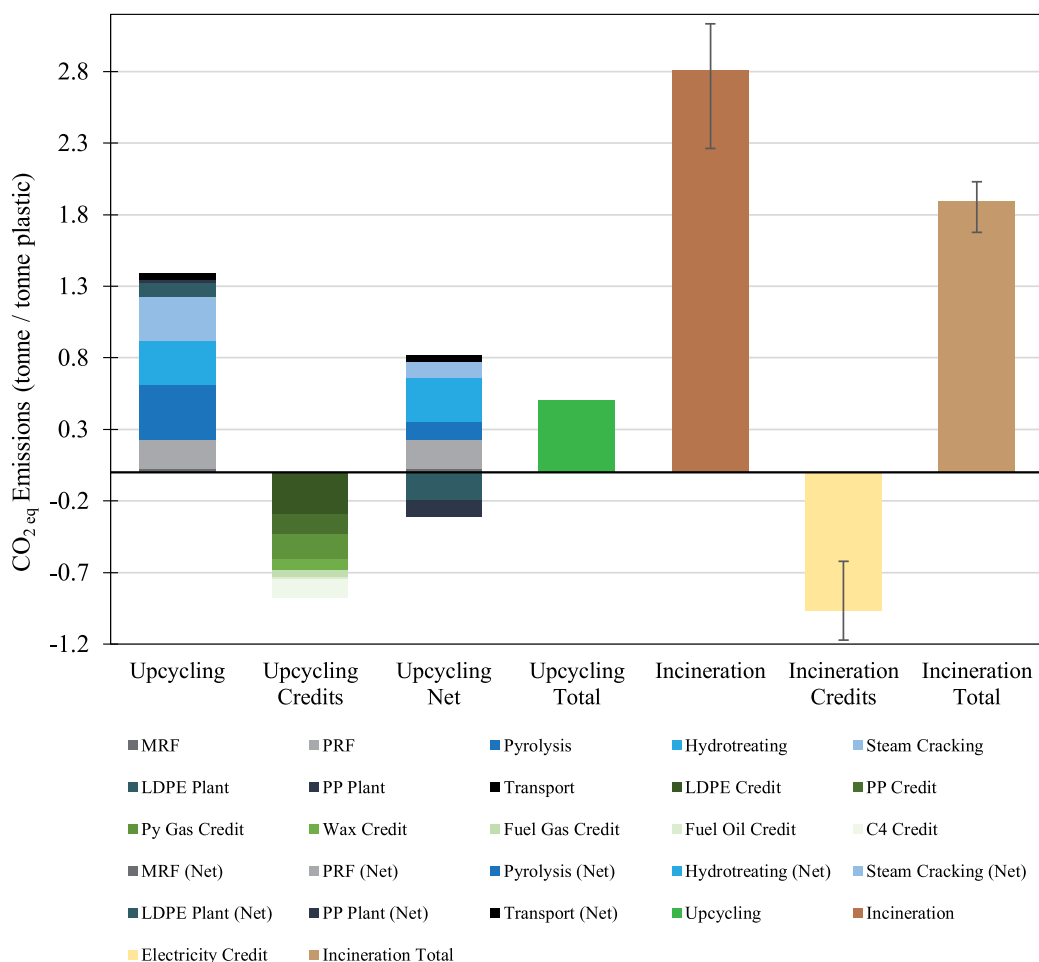


Fig. 8. Steps of upcycling contribute unevenly to GHG emissions, with pyrolysis and steam cracking (including hydrotreatment) creating the highest impacts. Still, the total emissions of upcycling (including MRF operation) is less than that of WtE (with error bars to represent the range of emissions estimates from combusting different types of plastic). However, crediting upcycling for producing polymers and chemicals that would otherwise necessitate additional fossil fuel use makes the comparison significantly more favorable. When these credits are applied to their respective technologies, some upcycling steps have net negative emissions (particularly the two polymerization steps). See Table 1 for EcoInvent LCA Database product identities used for emissions crediting.

and net impacts with error bars to represent a range of values representing different plastic waste stream compositions. Previous carbon footprint work conducted within the LCA framework has demonstrated that pyrolysis processes have lower global warming impacts than incineration technologies for mixed plastic waste (Vlasopoulos et al., 2023; Hermanns et al., 2023), but this analysis suggests that the impacts of the entire upcycling process (with or without byproduct credits) are also below the incineration benchmark range.

4. Discussion

Previous work has explored the steps of upcycling and modeled its implementation at smaller scales. Our analysis demonstrates that the chemical upcycling of waste plastics is economically viable on the scale of the entire continental US, while enabling a significant degree of circularity with regard to plastic waste production. Though processing costs of upcycling exceed those of existing methods, the value of its products are also correspondingly much greater. Furthermore, upcycling compares favorably with existing waste management systems, presenting significant reductions in land use (relative to landfills) and GHG emissions (relative to incineration or waste-to-energy).

Now that a basic economic picture of upcycling on the national scale has been established, future work could explore some of the simplifying assumptions of our analysis. Our infrastructure design represents the potential of upcycling for residential waste plastic: even though the US

already has a limited network of facilities such as MRFs, the arrangement of technologies in this study includes no existing facilities, which allows for simplifications in our analysis. Incorporation of these existing facilities could change minute details of the product transportation and facility placement patterns we observe, but is unlikely to impact our overall conclusion on the viability of this national system. Furthermore, the economic value created by the chemical reformation of polymers completely covers the capital costs of this potential network, so if existing facilities were included, total required investments would decrease, which could increase the value that propagates back to residential waste providers. However, we assume that model MRFs, with cost and yield factors based off of existing facilities, are able to sort all mixed plastic wastes, which may be beyond the capabilities of certain existing US MRFs.

Our infrastructure design is also structured based on a uniform distribution of waste plastic from only residential sources. The composition of real recyclable streams, and distribution of different types of plastic, can vary by location and socio-economic factors, providing another opportunity for the incorporation of region-specific composition data to refine the accuracy of our model results. If other plastic waste sources were to be included in the scope of our analysis, total required processing capacity (and therefore capital costs) could increase, but the processing capacity of this solution network is not completely exhausted.

The pyrolysis-based upcycling pathway modeled in this work is just one possible strategy for deriving value and circularity from mixed

plastic wastes. Future work can explore comparisons between upcycling pathways. Leveraging the versatility of the modeling framework outlined in this work, future analysis could focus on the competing priorities of various pathways, including their economic potential, environmental performance, and emphasis on plastic-to-plastic circularity under different cases of consumer behavior and policy. Our model involves many different stakeholders (including residents, technologies, and markets), but who would manage these technologies in practical implementations also deserves further consideration.

Likewise, future work should also explore the policy implications of upcycling as a waste management strategy. As demonstrated in this study, the potential of upcycling on the national scale represents a seismic shift in technologies, economics, and attitudes toward plastic recycling. Traditional mechanical recycling, though inherently more limited than upcycling, is an established pathway with a clear role to continue processing certain types of plastics in the future. To better understand the opportunities and obstacles for upcycling as part of an emerging national recycling strategy, we must not only refine our modeling results, but also look to the current regulatory environment for plastic recycling, engage with environmental, industry, and government stakeholders, and finally explore how upcycling would affect the state of environmental justice in plastic production and waste, in national and global contexts.

CRediT authorship contribution statement

Evan D. Erickson: Writing – original draft, Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Philip A. Tominac:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Jiaze Ma:** Writing – review & editing, Visualization, Methodology, Formal analysis, Data curation. **Horacio Aguirre-Villegas:** Writing – review & editing, Formal analysis, Data curation. **Victor M. Zavala:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

A link to the code and data was provided in the draft.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.compchemeng.2024.108800>.

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